Charge and the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain charge interactions and trace an electric car's energy transfers.

SUCCESS CRITERIA - I can...

1. Predict attraction or repulsion from charge signs.
2. Distinguish charge from energy.
3. Trace energy from a battery to moving wheels.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "charge" and "circuit" to explain your answer. Sentence starters are optional.

charge: a property responsible for electric interactions

circuit: a connected arrangement of electrical components

motor: a device converting electrical energy to mechanical output

01 Warm-up: bring back an earlier idea

An ideal blackbody has peak wavelength **966.7 nm**.
1. Use $\lambda_{max}\,T$=**2.90×10^{6} nm·K**.
2. **Estimate temperature**.

Word help: blackbody = an ideal absorber/emitter used in a thermal-radiation model

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_{electrical} = E_{mechanical} + E_{thermal}$ + other transfers. Energy accounting for a motor with no net energy storage. A battery transfers energy; charge already exists in the wires.

Worked reasoning: Rubbing can transfer electrons between materials. If one isolated object gains charge **-3** units and another loses the same electrons, their charge changes add to zero.

Your turn: Draw before-and-after charge accounts for both objects. Distinguish transfer from creation of charge.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Label charges and separation.

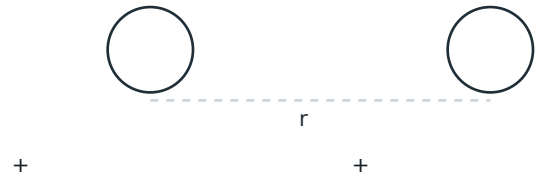
Object A Object B

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw force arrows for two positive charges, then for a positive and a negative charge.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

During a small classroom car test, the battery transfers **120 J** and the car gains **30 J** of translational kinetic energy. Its height stays fixed. **Complete the energy account**, find energy in other forms and **explain why the circuit's charge** is not used up.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says the motor uses up the current so less returns to the battery. **Explain the mistake** for a steady series circuit.

Sentence starter: The step I would change is ... . I would change it because I found ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Three instantaneous displacements are **+4 mm**, **-6 mm** and **-8 mm**. **Find their sum** and **explain why one displacement value** does not find total wave energy. **Show the signed contributions** and their sum.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

07 Exit check

How can an initially neutral object be attracted to a charged object without having a net charge?

Sentence starter: I notice ... . This supports ... because ... .



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

An initially neutral pair separates with charges **+8 nC** and **-8 nC**. One then shares all its net charge equally with an identical initially neutral conductor. **Find the three final charges**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3